

SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

**COURSE NAME: Advance Wireless Communication Systems /
5 G / 6G Communication Systems**
COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 2 - *Analyse LTE and 5G wireless network architectures, including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities. (BL4)*

Assessment Tool Number	: CO2_AT2
Assessment Tool Weightage	: 35%
Assessment Tool	: Simulation Task
Duration	: Self Learning
Marks	: 100 Marks
Type of Assessment Conduction	: At Home
Evaluation Type	: Online Evaluation(GitHub)
Result Generation	: Students successfully simulate the assigned wireless communication system, analyze the obtained results, and interpret the performance based on relevant technical parameters.

Topic Covered : LTE/LTE-Advanced architecture, protocol stack, carrier aggregation, higher-order MIMO, beamforming, and beamsteering techniques. 5G NR air interface, gNodeB and RAN architecture, RRC procedures, 5G Core, Service-Based Architecture, network slicing, AMF, SMF, UPF, and control/user plane separation.

Conduction of Assessment Procedure:

- Assessment will be carried out through rubric-based evaluation, where students complete the assigned task.
- Simulation will be done in MATLAB/Simulink for the given specification and interpret the results.

- TWO questions will be assigned to students each carrying 10 marks and they are required to collaboratively discuss the given problem and present their responses with clear justification and submit the document in GitHub Platform.
1. Simulation of LTE protocol stack delay analysis under uplink traffic, downlink traffic, packet size, retransmission count, and channel quality
 2. Simulation of edge computing delay under local processing, cloud processing, latency, bandwidth, and task size
-

Code of Conduct:

I R.harshavardha / 192412308 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Simulation Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	20 %	Demonstrates strong understanding of underlying concepts in the simulation(10)	Shows good understanding with minor gaps(8)	Partial understanding; some misconceptions (6)	Lacks understanding of key concepts(4)
Model Design	25 %	Develops accurate, well-structured simulation/model independently (12.5)	Develops correct model with minor errors(10)	Model is incomplete or requires guidance(8)	Unable to develop a proper model(6)
Tool Usage	20 %	Uses simulation tools effectively with correct setup and execution(10)	Uses tools with minor errors or guidance(8)	Limited ability to use tools properly(6)	Unable to use simulation tools(4)
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results(10)	Analysis is reasonable with minor gaps(8)	Superficial analysis; limited interpretation(6)	Unable to analyze or interpret results(4)
Documentation	15%	Presents results clearly with proper documentation and structured reporting(7.5)	Documentation is adequate with minor issues(5)	Poorly organized or incomplete documentation (3)	No proper documentation or presentation (1)

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Q.NO	Conceptual Understanding	Model Design	Tool Usage	Analysis of Results	Documentation	TOTAL
1						
2						
TOTAL						

1. LTE Protocol Stack Delay Analysis

```
clc; clear; close all;
```

```
% Parameters
```

```
N = 100; % Number of sessions
```

```
sessionCount = (1:N)';
```

```
AMF_load = randi([20 95], N, 1); % AMF Load in %
```

```
SMF_load = randi([15 90], N, 1); % SMF Load in %
```

```
UPF_throughput = randi([100 1000], N, 1); % UPF Throughput in Mbps
```

```
% Latency calculation based on Core Network Load and Throughput
```

```
baseLatency = 5; % Base processing latency in ms
```

```
AMF_delay = (AMF_load / 100) * 12;
```

```
SMF_delay = (SMF_load / 100) * 10;
```

```
UPF_delay = (1000 ./ UPF_throughput) * 15;
```

```
totalLatency = baseLatency + AMF_delay + SMF_delay + UPF_delay;
```

```
% Output Summary to Command Window
```

```
fprintf('Average 5G SBA Latency = %.2f ms\n', mean(totalLatency));
```

```
fprintf('Minimum 5G SBA Latency = %.2f ms\n', min(totalLatency));
```

```
fprintf('Maximum 5G SBA Latency = %.2f ms\n', max(totalLatency));
```

```
ResultTable = table(sessionCount, AMF_load, SMF_load, UPF_throughput,  
totalLatency, ...
```

```
    'VariableNames', {'SessionID', 'AMF_Load_Pct', 'SMF_Load_Pct',  
'UPF_Throughput_Mbps', 'Latency_ms'});
```

```
disp(ResultTable(1:10, :)); % Display top 10 results
```

```
% Plotting Results
```

```
figure;
```

```
subplot(2,2,1);
```

```
scatter(AMF_load, totalLatency, 'filled', 'b'); grid on;
```

```
xlabel('AMF Load (%)'); ylabel('Latency (ms)'); title('AMF Load vs  
Latency');
```

```
subplot(2,2,2);
```

```
scatter(SMF_load, totalLatency, 'filled', 'g'); grid on;
```

```
xlabel('SMF Load (%)'); ylabel('Latency (ms)'); title('SMF Load vs  
Latency');
```

```
subplot(2,2,3);
```

```
plot(UPF_throughput, totalLatency, 'r*'); grid on;
```

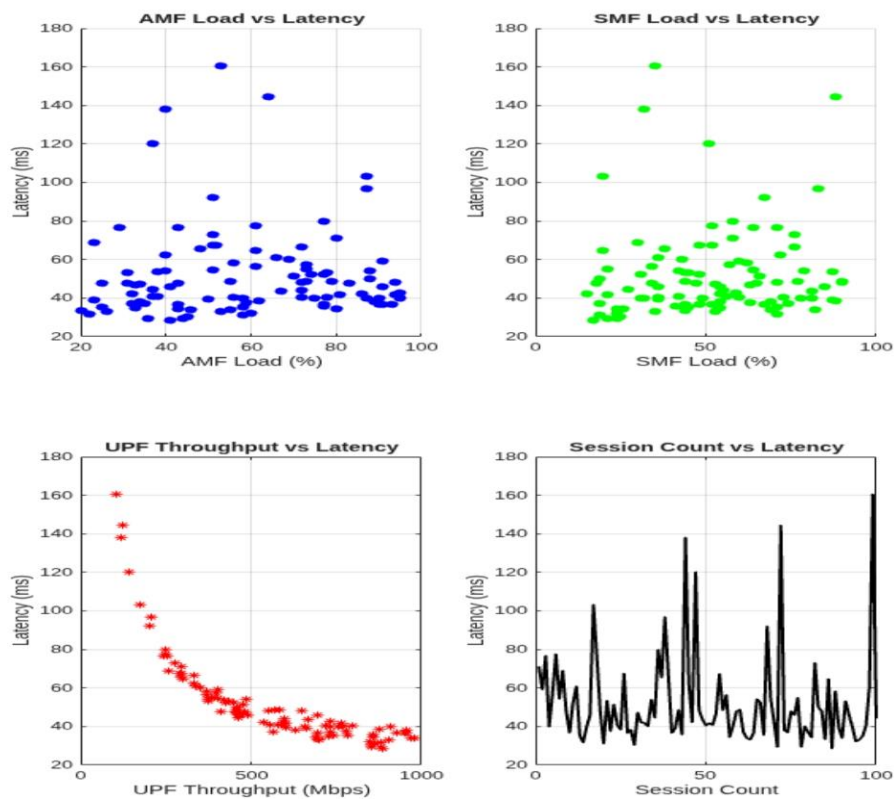
```
xlabel('UPF Throughput (Mbps)'); ylabel('Latency (ms)'); title('UPF  
Throughput vs Latency');
```

```
subplot(2,2,4);
```

```
plot(sessionCount, totalLatency, 'k-', 'LineWidth', 1.5); grid on;
```

```
xlabel('Session Count'); ylabel('Latency (ms)'); title('Session Count vs  
Latency')
```

OUTPUT



2. Edge Computing Delay Simulation

```
clc; clear; close all;
```

```
% Simulation Parameters
```

```
SNR_dB = 0:2:30; % SNR range in dB
```

```
SNR_linear = 10.^(SNR_dB / 10);
```

```
B = 20e6; % Bandwidth = 20 MHz
```

```
multiplexingOrder = [1, 2, 4, 8]; % Antenna Configurations (N x N MIMO)
```

```
figure;
```

```
hold on; grid on;
```

```
colors = ['r', 'g', 'b', 'm'];
```

```
% Capacity Calculation over Rayleigh Fading
```

```
for idx = 1:length(multiplexingOrder)
```

```
    Nt = multiplexingOrder(idx); % Tx Antennas
```

```
    Nr = Nt; % Rx Antennas
```

```
    C = zeros(1, length(SNR_dB));
```

```
    numTrials = 500; % Rayleigh Channel realizations
```



```

for k = 1:length(SNR_linear)
    snr = SNR_linear(k);
    cap_sum = 0;
    for t = 1:numTrials
        H = (randn(Nr, Nt) + 1i*randn(Nr, Nt)) / sqrt(2); % Rayleigh
Channel
        cap_sum = cap_sum + real(log2(det(eye(Nr) + (snr / Nt) * (H *
H')))));
    end
    C(k) = (cap_sum / numTrials) * (B / 1e6); % Capacity in Mbps
end

plot(SNR_dB, C, [colors(idx) '-o'], 'LineWidth', 1.5, 'DisplayName',
sprintf('%dx%d MIMO', Nt, Nr));
end

xlabel('SNR (dB)');
ylabel('Channel Capacity (Mbps)');
title('MIMO Capacity vs SNR under Rayleigh Fading');
legend('Location', 'NorthWest');

% Print Summary Results
fprintf('Maximum Capacity achieved at 30 dB SNR:\n');

```

```

for idx = 1:length(multiplexingOrder)

    fprintf('%dx%d MIMO: %.2f Mbps\n', multiplexingOrder(idx),
multiplexingOrder(idx), C(end));

end

```

OUTPUT

← **Figures** SHOW DATA CURSOR ⋮

figure 1: clc; clear; close all; % Simulation Parameters SNR_dB =...

